



Figure 3: Spinnaker platform

Ansible

URL: ansible.com

Key features

- *Smart automation:* AI-driven playbooks that adapt to changing conditions in the environment, making automation more robust and resilient.
- *Anomaly detection:* Identifies unusual patterns in system configurations and performance metrics.
- *Predictive maintenance:* Forecasts potential system failures and triggers pre-emptive maintenance tasks.

Use cases

- Automating complex workflows with adaptive playbooks.
- Proactive system maintenance and issue prevention.

Puppet

URL: puppet.com

Key features

- *Intelligent automation:* AI enhances automation scripts to be more adaptive and context aware.
- *Self-healing systems:* Automatically identifies and remediates configuration drifts and system issues.
- *Performance insights:* Provides in-depth analysis of infrastructure

performance and suggests optimisations.

Use cases

- Self-healing infrastructure for improved uptime.
- Enhanced configuration management with predictive insights.

Apache Airflow

URL: airflow.apache.org/

Key features

- Workflow automation and scheduling platform.
- Supports creating, managing, and monitoring complex workflows.
- Extensible with plugins for various integrations.

Use cases

- Orchestrating machine learning workflows.
- Automating data pipelines and model training.

Prometheus

URL: prometheus.io

Key features

- Open source monitoring and alerting toolkit.
- Provides powerful query language and time-series database.
- Integrates with Grafana for

visualisation.

Use cases

- Monitoring machine learning models in production.
 - Collecting and analysing metrics from AI applications.
- These tools leverage AI to bring a new level of intelligence and automation to DevOps processes, significantly enhancing efficiency, reliability, and responsiveness.

Research dimensions in AIOps

The confluence of AI and DevOps, known as AIOps, heralds a transformative era in IT operations by amalgamating advanced AI techniques with the established practices of DevOps. A pivotal dimension in AIOps research is the application of machine learning algorithms to anomaly detection and predictive analytics. These algorithms scrutinise vast datasets encompassing system logs, performance metrics, and user interactions to discern patterns that might elude human operators. By predicting potential system failures or performance bottlenecks, AIOps enables pre-emptive remediation measures, thereby minimising downtime and enhancing system reliability.

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